



The SoReMo Initiative



ILLINOIS INSTITUTE OF TECHNOLOGY

Client flow in food pantries: check-in presentation no. 1

Leonardo Marciaga

Department of Applied Mathematics
January 29, 2026

Project recap

Food pantries face several operational challenges:

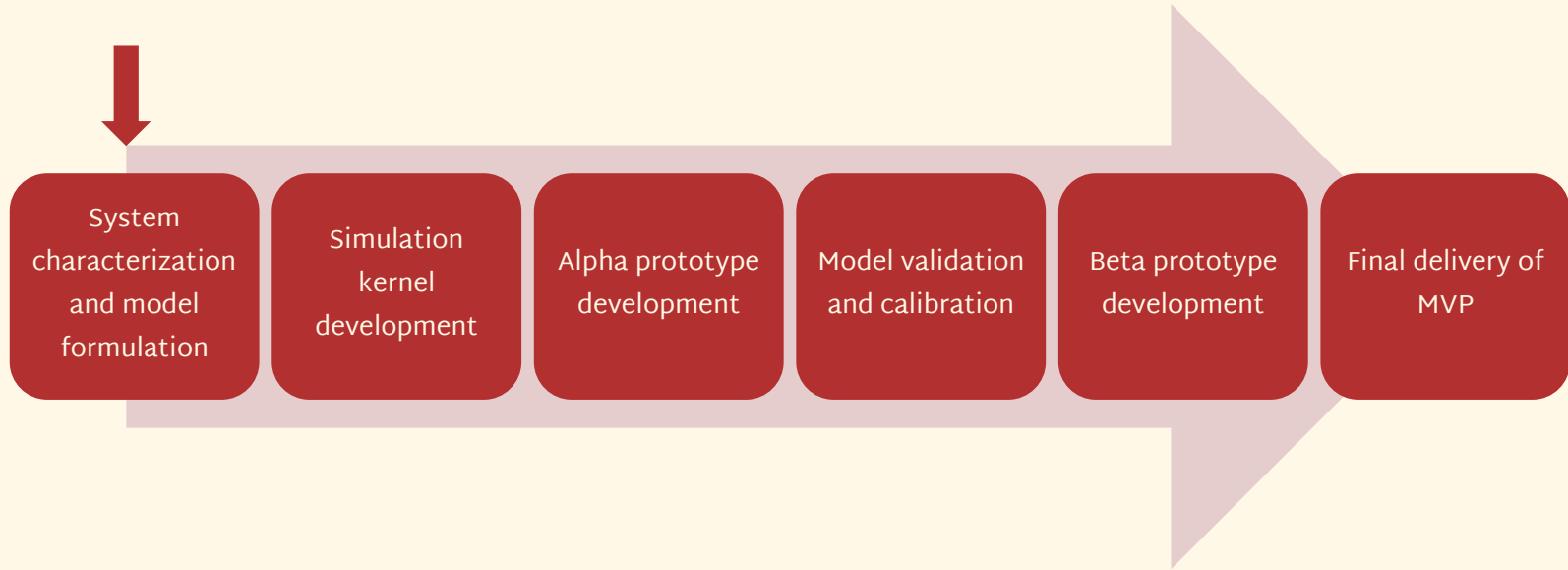
Uncertainty in
demand and
volunteer staff

System throughput
vs. service
autonomy

Data scarcity

Goal: build a tool using discrete-event simulation to optimize client flow in a local food pantry.

Roadmap



Current milestone: process scope definition and documentation.

Overview of food pantry processes

Opening process

- Determining items to be served, arranging items in shelves/containers, and volunteer training.

Shopping process

- Letting clients into the building, check-in and volunteer-guided shopping.

Volunteers are assigned tasks **dynamically**.

Main challenges

Dynamic task
allocation

Varying levels of
volunteer supply
and experience

Need for daily
volunteer training

Opening/shopping
process overlap

Proposed problem scope: managing volunteer allocation under low staff supply, with constraints on the volunteers' level of experience, focused on the opening process.

Next steps

Standardized process
documentation

Literature search

Process map
formulation and
validation

Mathematical model
formulation

System architecture
specification

Thanks!